October 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **"Machine learning in battery management systems: a two-tier scoping review of 799 eligible studies with in-depth classification and validation audit"** for consideration as a Review Paper in *IEEE Transactions on Transportation Electrification* (12-page limit including references, per the TTE information for authors).

The manuscript links every BMS machine-learning function to the requirement an electric-vehicle safety standard actually places on it — UN GTR 20's thermal-propagation early-warning window (Amendment 1, endorsed for WP.29 adoption in late 2026), GB 38031-2025's no-fire/no-explosion thermal-diffusion tests (effective July 2026, superseding GB 38031-2020), and ISO 26262:2018 extended to learned components by ISO/PAS 8800:2024, with ISO 21448 (SOTIF) reasoning for capability limitations. The review's measured finding is that the classified literature rests predominantly at cell level and in the laboratory (36 of 88 primary papers on the authors' own cells; real-vehicle data in 10; online validation in 8), one integration level below the pack- and vehicle-level questions these standards pose. That gap — and the reporting checklist proposed to close it — is written for the reader deciding how ML fits into a certifiable EV traction system.

The evidence base is two-tier. A rule-based eligible corpus of 799 peer-reviewed journal articles (798 dated 2020–2026 plus one early-access record stamped 2027) carries the measured bibliometrics; a curated 103-study classified subset (98 from the pool plus 5 individually retrieved seeds) carries the in-depth coding, with a 30-paper blind inter-coder reliability study, a per-paper protocol audit, and a label-noise sensitivity analysis. Within the subset, only 8 of the 88 primary-application papers demonstrate online or fleet-level operation, 17 rely on within-cell or k-fold splits, and 23 rest on three public benchmarks while 36 rest on private laboratory cells; the physics-informed validation contrast is directionally consistent under both reported definitions but not statistically significant (36% vs. 52%, $p = 0.32$; 59% vs. 74%, $p = 0.19$). All screening provenance and analysis code are released in a Zenodo deposit (DOI: 10.5281/zenodo.23436202); every count in the paper can be regenerated from the deposit.

All validation-practice rates and reliability statistics are reported for the classified subset and are not generalized to the field; caveats about the capped Crossref retrieval (including a documented January–May 2024 retrieval hole and publisher tilt toward energy-storage venues), the curated subset, and the non-significance of the physics-informed contrast are stated in the manuscript rather than left to the reader. The review was not prospectively registered; this is stated explicitly.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the submission and declare no conflicts of interest. Writing-support tools (ChatGPT, Grammarly, QuillBot) were used for grammar, readability, and sentence-restructuring support only; no generative-AI tool wrote manuscript prose, and all technical content, analysis, and interpretations are the authors' own. The complete classified corpus, the screening provenance, and all analysis scripts are publicly

available at https://github.com/touhidsiddiqueeraj-bit/-structured-scoping-review-and-classifica

Thank you for your consideration.

Sincerely,

Hussain Touhid Siddiquee (corresponding author)
On behalf of all authors
Leading University, Sylhet, Bangladesh